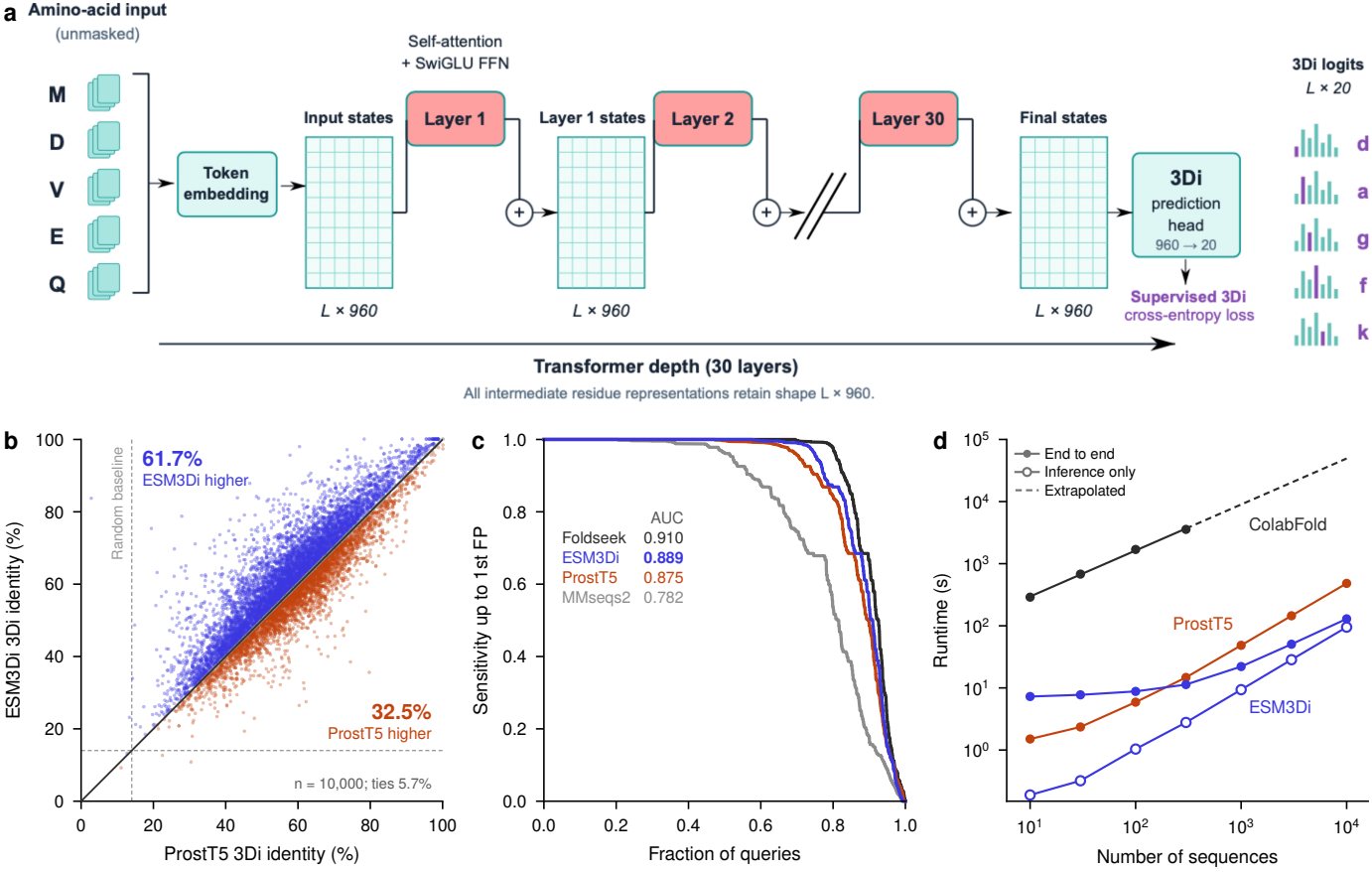

a
Amino-acid input
(unmasked)
M
D
V
E
Q
Token embedding
Input states
L × 960
Self-attention
+ SwiGLU FFN
Layer 1
Layer 1 states
L × 960
Layer 2
Layer 30
Final states
L × 960
3Di
prediction head
960 → 20
Supervised 3Di
cross-entropy loss
3Di logits
L × 20
d
a
g
f
k
Transformer depth (30 layers)
All intermediate residue representations retain shape L × 960.
b
61.7%
ESM3Di higher
Random baseline
32.5%
ProstT5 higher
n = 10,000; ties 5.7%
ESM3Di 3Di identity (%)
ProstT5 3Di identity (%)
c
AUC
Foldseek 0.910
ESM3Di 0.889
ProstT5 0.875
MMseqs2 0.782
Sensitivity up to 1st FP
Fraction of queries
d
End to end
Inference only
Extrapolated
ColabFold
ProstT5
ESM3Di
Runtime (s)
Number of sequences